

MIT-BIH Arrhythmia Heartbeat Classification

Logistic Regression, Random Forest and CNN Approaches to Four-Class Heartbeat Classification

Md Soad Solaiman

Task 1: Data Preparation and Exploration

1.1 Dataset Description

This project uses the MIT-BIH Arrhythmia Database, developed by Moody and Mark (2001a) and distributed through PhysioNet (Pollard et al., 2026). The database contains 48 half-hour ECG recordings from 47 subjects. Each recording captures two channels at a sampling frequency of 360 Hz, with 11-bit resolution over a 10 mV range. Every heartbeat was annotated by cardiologists with reliable label for each beat. The 48 recordings were not chosen entirely at random. Twenty-three came from a random sample of a larger pool and the rest from rarer arrhythmias. This combination of scale, expert labelling and rhythm diversity makes the dataset well suited to heartbeat classification.

1.2 Class Definition

Each heartbeat in MIT-BIH carries a raw annotation symbol assigned by the reviewing cardiologist. These symbols were mapped onto five AAMI-style groups, following the scheme used by de Chazal, O'Dwyer and Reilly (2004). N represents normal-type beats. S represents supraventricular ectopic beats, originating in the heart's upper chambers. V represents ventricular ectopic beats, originating in the lower chambers. F represents fusion beats, combining normal and abnormal activity. Q represents paced or unclassifiable beats. Q was excluded before modelling as 99% of its data is coming from pacemaker patient.

1.3 Exploratory Data Analysis

The exploratory analysis covered seven areas. Each one is reported below with what was examined, what was found, and why the finding matters for the modelling decisions.

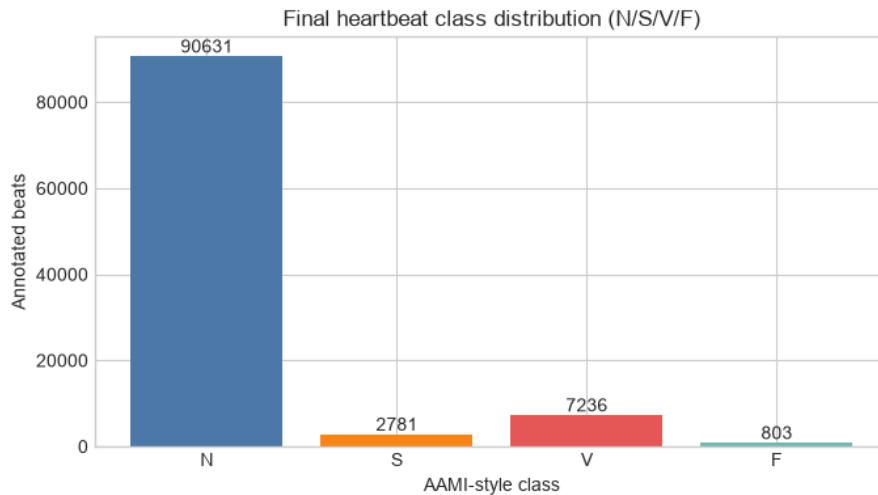


Figure 1. Final heartbeat class distribution across all 48 recordings, after AAMI mapping.

Overall class distribution: Figure 1 shows N beats dominate. N makes up 82.77 percent. F makes up under one percent. This is a severe imbalance. A model predicting only N would

already be mostly correct, without learning anything useful. This is why per-class metrics are used throughout this report.

Record-level class distribution: Class balance also varies sharply by recording. Records 208 and 213 together hold 735 of the 803 F beats. Most other recordings hold very few, or none. Splitting individually could let a model learn these two patients, not genuine F morphology. Splitting by whole recording, per Section 1.4, removes this risk.

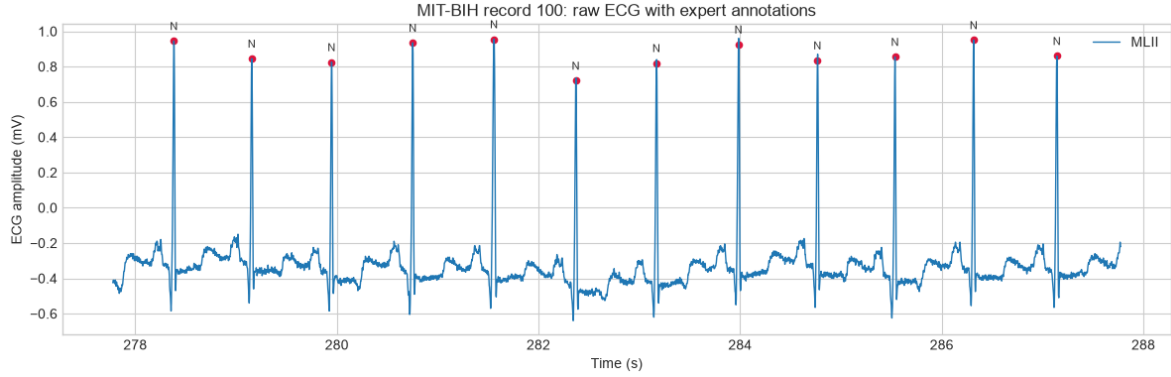


Figure 2. A ten-second segment of raw ECG signal from record 100, with each annotated beat marked at its R peak.

ECG waveform exploration: Figure 2 shows a short segment of raw signal from one recording, with each annotated beat marked at its R peak. Every marked beat aligns with a sharp upward spike, the expected shape of a normal QRS complex. This confirms the annotations correspond to real, visible heartbeat activity, and that the signal is clean enough to support segmentation.

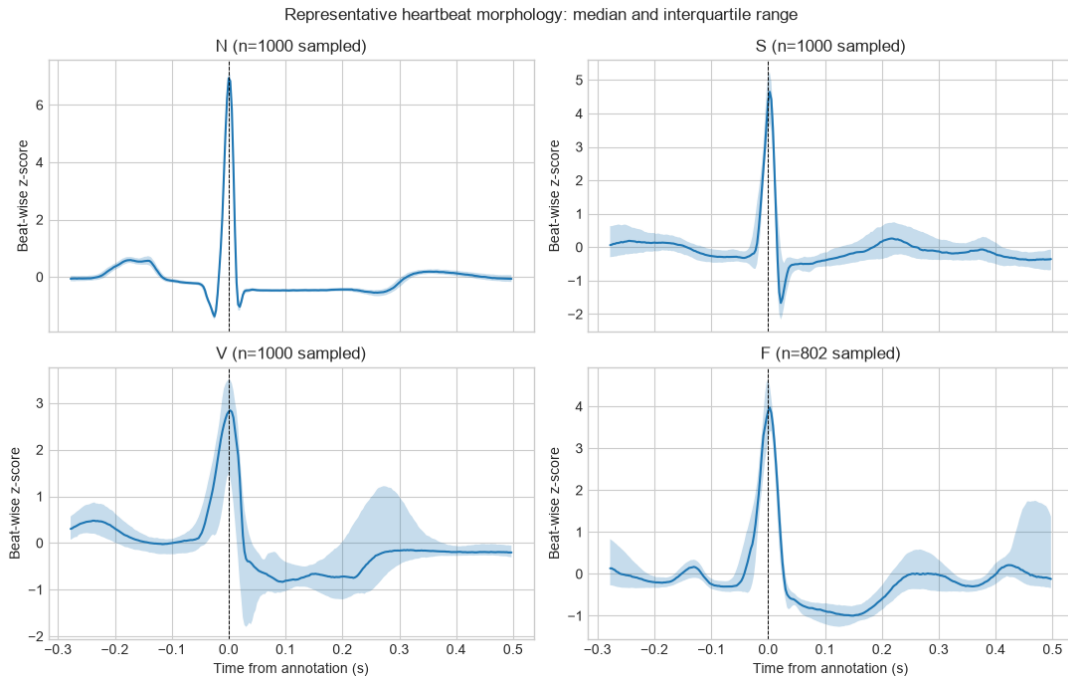


Figure 3. Representative heartbeat morphology by class.

Representative morphology by class: Figure 3 shows the median waveform and interquartile range per class. N shows a tall, narrow peak. V shows a wider peak with a broader interquartile

band, consistent with irregular morphology. S resembles N closely. F sits between N and V. Considerable within-class variation means median shape alone cannot guarantee separability, motivating the engineered features in Section 1.6.

1.4 Data Split

Data was split by whole recording rather than by individual beat. This prevents beats from the same patient appearing in more than one partition, which would otherwise inflate performance. Recordings were assigned to training, validation or test sets. The validation set supported model and hyperparameter selection. The test set remained untouched until final evaluation. The split follows the 67-16.7-16.7 percentile.

Table 1: Number of recordings and heartbeats assigned to each partition of the patient-independent split.

Split	Records	N	S	V	F	Total beats
Train	32	63,622	2,037	5,025	410	71,094
Validation	8	13,630	310	911	367	15,218
Test	8	13,379	434	1,300	26	15,139

1.5 Preprocessing

Filtering: Raw signals contain noise from breathing, muscle movement and recording equipment. A third-order, zero-phase Butterworth band-pass filter from 0.5 to 40 Hz. It removes slow baseline wander and fast noise while leaving the heartbeat shape intact.

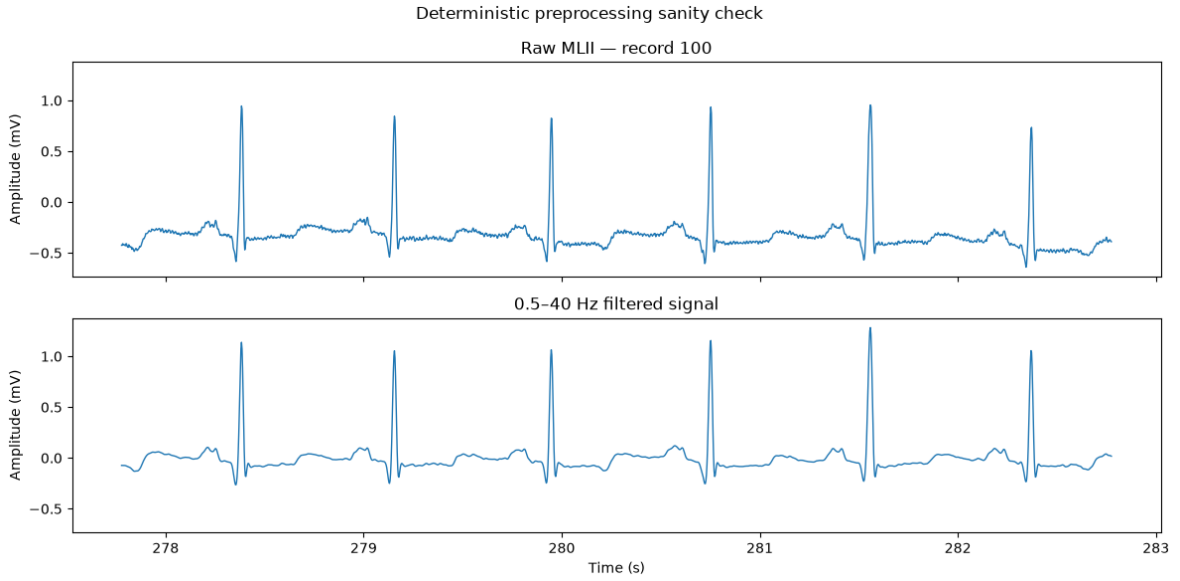


Figure 4. A representative ECG segment before and after filtering.

Segmentation: Each heartbeat was cut into a fixed 280-sample window around its R peak. This covered 100 samples before and 180 after, equal to 0.278 and 0.500 seconds. The window is wide enough to capture the beat without overlapping the next one. Edge beats without a full window were excluded.

Normalization: Each window was rescaled using only its own mean and standard deviation. Validation or test data haven’t entered this step.

Missing values and scaling: Missing edge RR values were filled using the recording's own median values rather than discarded. The classical models' scaler was fitted on training data only.

1.6 Feature Extraction

Rather than feeding raw signal numbers directly into the simpler models, eighteen descriptive measurements were calculated for every heartbeat. These are called features. They fall into three groups. Five features describe timing and rhythm, such as the gap to the previous and next heartbeat. Nine features describe the shape and energy of the waveform, such as its highest point, lowest point, and overall spread. The remaining four describe how quickly the signal changes and how complex its shape looks.

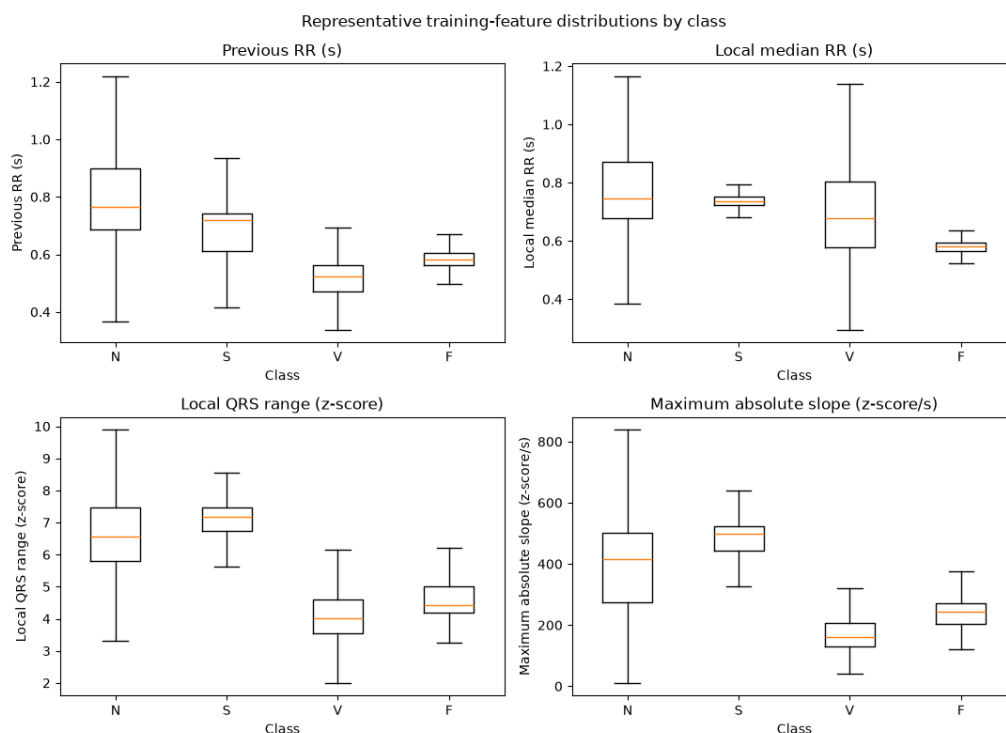


Figure 5. How four of these measurements differ across the four heartbeat types.

As Figure 5 shows, even a handful of these measurements already look visibly different across heartbeat types. It suggests these features carry real information a model could learn from.

Task 2: Model Selection and Training

Three models were trained and compared. They are Logistic Regression, Random Forest and a Convolutional Neural Network (CNN). Logistic Regression is a linear model. Random Forest and CNN are non-linear models. Comparing all three shows what nonlinearity adds and what learned features add. Each model below follows the same steps: input, model characteristics, training, hyperparameter tuning, validation, final model selection, test evaluation, and interpretation.

2.1 Logistic Regression

Input

Logistic Regression uses the eighteen engineered features from Section 1.6. These cover rhythm, waveform statistics, and morphology.

Model characteristics

Logistic Regression is a linear model. For each class, it computes a weighted sum of the eighteen features. This weighted sum becomes a score. The four scores are converted into probabilities. The class with the highest probability is the prediction.

$$score(k) = w0 + w1 \times feature1 + w2 \times feature2 + \dots + w18 \times feature18$$

Training

Before training, the eighteen features were rescaled. The scaler was fitted using training data only. Class weights were set to balanced, giving more weight to errors on the rare classes.

Hyperparameter tuning

One setting was tuned is C which controls how much the model relies on any single feature. Three values were tried, each evaluated on validation data using macro F1.

Table 2: Logistic Regression candidate values of C, evaluated on validation data.

C	Validation Accuracy	Validation Macro F1
0.1	0.8082	0.5548
1.0	0.8096	0.5627
10.0	0.8109	0.5662

Final model selection

C = 10 was selected. It gave the highest validation macro F1 of the three values tried. The test set was not used at this stage.

Test evaluation

Table 3: Logistic Regression validation and test performance.

Dataset	Accuracy	Macro Precision	Macro Recall	Macro F1
Validation	0.8109	0.5362	0.7267	0.5662
Test	0.6903	0.4936	0.6115	0.4555

Table 4: Logistic Regression per-class test performance (15,133 beats).

Class	Precision	Recall	F1	Support
N	0.9862	0.6768	0.8027	13,374
S	0.1137	0.9309	0.2027	434
V	0.8729	0.7614	0.8133	1,299
F	0.0016	0.0769	0.0031	26

Interpretation

Macro F1 fell from 0.5662 on validation to 0.4555 on test. For S beats, recall is high, at 0.931, so the model catches almost every true S beat. The model is eager to flag S and V as well. But not for N whereas its dataset is big. F is just like not working as its dataset very low making the model to overfit.

2.2 Random Forest

Input

Random Forest uses the same eighteen features as Logistic Regression. No scaling is needed for tree-based models.

Model characteristics

Random Forest is a non-linear model. It builds many decision trees. Each is trained on a random sample of the training beats. A single tree works like a flowchart, asking a yes-or-no question about one feature at each step. The final prediction is the class most trees agree on.

Training

Class weights were set to balanced subsample, adjusting for class imbalance separately within each tree’s bootstrap sample.

Hyperparameter tuning

Three configurations were tried, each evaluated on validation data using macro F1.

Table 5: Random Forest candidate configurations, evaluated on validation data.

Configuration	Trees	Max Depth	Min Leaf	Max Features	Val. Macro F1
RF1	200	None	1	sqrt	0.4788
RF2	200	20	2	sqrt	0.5348
RF3	300	20	5	0.7	0.6166

Final model selection

RF3 was selected for its macro F1, not its accuracy. The final Random Forest has 300 trees, a maximum depth of 20, a minimum leaf size of 5, and considers 70 percent of features at each split. The test set was not used at this stage.

Test evaluation

Table 6: Random Forest validation and test performance.

Dataset	Accuracy	Macro Precision	Macro Recall	Macro F1
Validation	0.9165	0.6001	0.6560	0.6166
Test	0.8922	0.5044	0.5003	0.4979

Table 7: Random Forest per-class test performance (15,133 beats).

Class	Precision	Recall	F1	Support
N	0.9630	0.9154	0.9386	13,374
S	0.1182	0.1198	0.1190	434
V	0.9348	0.9276	0.9312	1,299
F	0.0014	0.0385	0.0028	26

Interpretation

Macro F1 fell from 0.6166 on validation to 0.4979 on test. Recall stays high for N and V, at 0.915 and 0.928. But its recall is surprisingly poor for S. It means it is efficient detecting N and V but not for S and F.

2.3 CNN

Input

The 280 sample raw waveform for each beat is the input of the CNN.

Model characteristics

The CNN is a non-linear model that learns its own representation from the waveform. A convolutional layer slides a small filter along the signal. It computes a weighted sum of nearby values at each position. Also, it produces a new signal that highlights one local pattern. The network has two convolutional layers with 32 and 64 filters. The parameters are 19,524 in total.

Training

Training used the Adam optimiser, a learning rate of 0.001, and a batch size of 256.

Hyperparameter tuning and validation

The CNN was not tuned with a grid search. Training ran for up to 15 epochs, with early stopping watching validation loss.

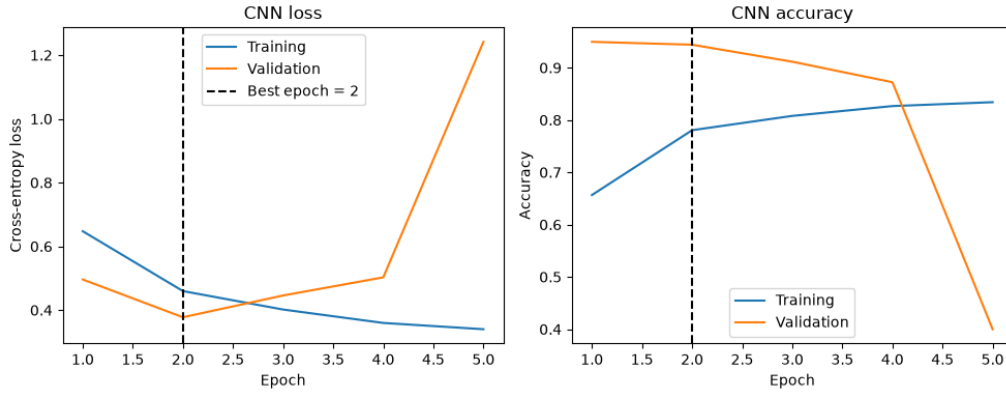


Figure 7. CNN training and validation loss and accuracy, by epoch.

Figure 7 shows performance across five epochs. Validation loss reached its lowest point at epoch 2, at 0.3776, then rose sharply to 1.2416 by epoch 5. Falling training loss alongside rising validation loss is the standard sign of overfitting. That’s why hyperparameter tuning is avoided here.

Final model selection

The checkpoint from epoch 2 was restored and used for evaluation, the point of lowest validation loss, not the final epoch’s weights.

Test evaluation

Table 8: CNN validation and test performance, at the restored epoch 2 checkpoint.

Dataset	Accuracy	Macro Precision	Macro Recall	Macro F1
Validation	0.9440	0.4644	0.4858	0.4749
Test	0.9522	0.4836	0.4570	0.4686

Table 9: CNN per-class test performance (15,133 beats).

Class	Precision	Recall	F1	Support
N	0.9526	0.9967	0.9741	13,374
S	0.0000	0.0000	0.0000	434
V	0.9818	0.8314	0.9004	1,299
F	0.0000	0.0000	0.0000	26

Interpretation

It's recall for N and V is really high. But unfortunately, CNN scored zero recall for both S and F on the test set. It suggests it had already failed to learn these two classes by validation. It's because, in front of its parameters, the dataset of S and F are very small making the overfit for them.

Task 3: Model Evaluation and Interpretation

The priority is to catch abnormal heartbeats, not simply to be correct overall. So, recall is treated as the most important metric throughout this section. Because in clinical setting, false alarm is less costly for a human life than a miss diagnosis. F1 is reported as a secondary metric, more relevant if the priority shifts toward reducing false alarms.

3.1 Overall Test Results

Table 10: Overall test results for all three models. Precision, recall and F1 are macro averages across all four heartbeat types.

Model	Accuracy	Precision	Recall	F1	AP
Logistic Regression	0.6903	0.4936	0.6115	0.4555	0.6304
Random Forest	0.8922	0.5044	0.5003	0.4979	0.5152
CNN	0.9522	0.4836	0.4570	0.4686	0.4710

Table 10 ranks the models differently depending on the metric used. Ranked by accuracy, the CNN comes first. But ranked by recall, Logistic Regression comes first. A model that misses fewer true abnormal beats is doing its core job better, even if its overall accuracy is lower. Random Forest has the highest F1, at 0.4979, which would matter more if the goal shifted toward reducing false alarms. The CNN's accuracy, at 0.9522, is driven almost entirely by the majority N class, not by its ability to detect abnormal beats.

3.2 Confusion Matrices

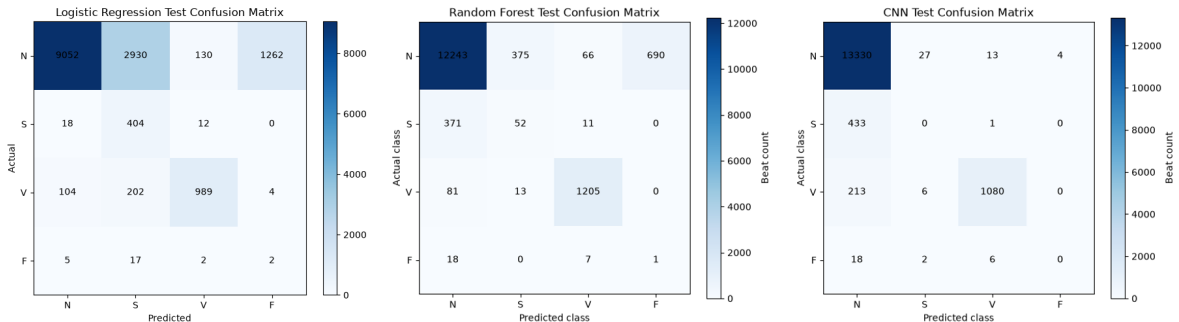


Figure 8. Confusion matrices for Logistic Regression, Random Forest and CNN (left to right).

Figure 8 shows a clear pattern. Logistic Regression misclassifies 4,322 true N beats as abnormal, a large number of false alarms. Random Forest reduces this to 1,131, a reduction of 3.8 times. The CNN is strongest on N, outperforming both other models. All three share one weakness: none detects F correctly. This is not a modelling failure. Only 26 F beats exist among the 15,133 test beats, too little evidence for any model to learn from.

3.3 Precision-Recall Curves and Average Precision

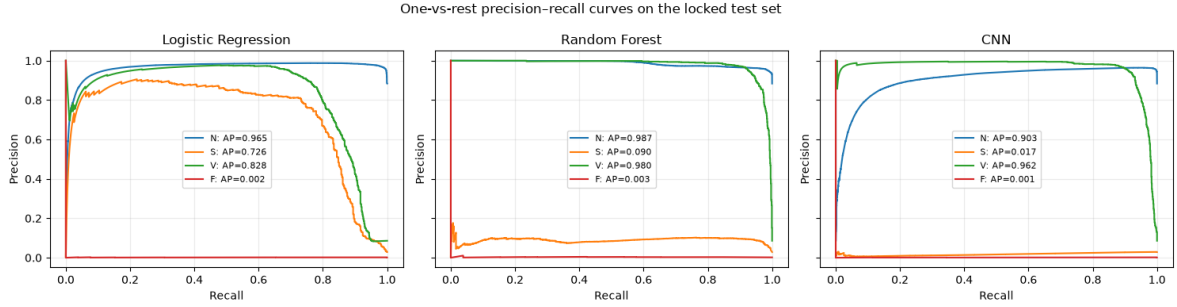


Figure 9. Precision-recall curves for each model, by heartbeat type, with Average Precision (AP) shown for each class.

Figure 9 makes the S and F difference clear. Logistic Regression's S curve stays reasonably high, reaching an AP of 0.7263. Random Forest's S curve is much weaker, at 0.0901. The CNN's is weaker still, at 0.0174. Every model's F curve collapse toward zero. It confirms that F is not learnable from this training set.

3.4 McNemar Comparison on Recall

McNemar's test compares two models on the same beats. It checks how often one model catches an abnormal beat the other misses. Since recall matters most here, the test only looks at true abnormal beats. A model succeeds if it predicts any abnormal class.

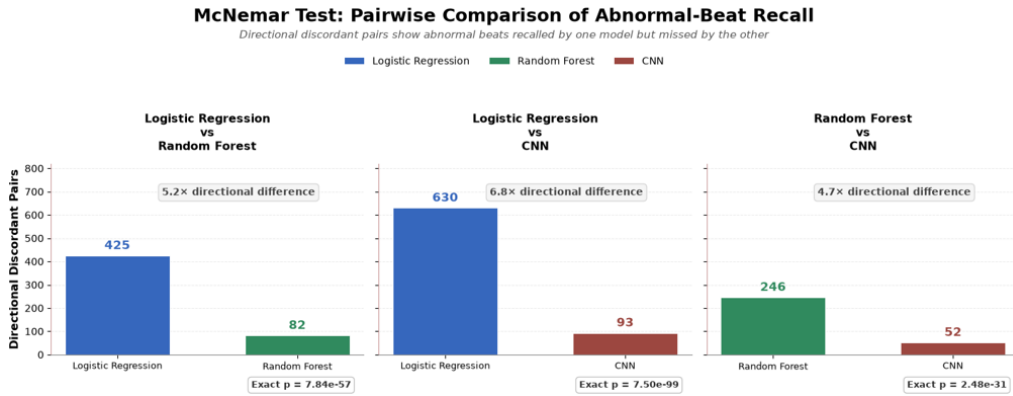


Figure 10. McNemar test comparing abnormal-beat recall, shown pairwise for all three models.

A p-value below 0.05 means the difference is statistically significant at the 95% confidence level. All three comparisons here fall far below that threshold. This means the differences are statistically significant on 95% confidence level.

Figure 10 shows the results. Logistic Regression beats Random Forest. It catches 425 abnormal beats Random Forest misses. Random Forest only catches 82 that Logistic Regression misses. This is significant at $p = 7.84e-57$.

Logistic Regression beats CNN even more clearly. It catches 630 abnormal beats CNN misses. CNN only catches 93 the other way. This is significant at $p = 7.50e-99$.

Random Forest also beats CNN. It catches 246 abnormal beats CNN misses, against just 52 the other way. This is significant at $p = 2.48e-31$.

Logistic Regression wins every comparison. It is statistically significantly better than both Random Forest and CNN at recalling abnormal beats. Based on this test, Logistic Regression is the best of the three models for recall.

3.5 Model Complexity and Its Effect on Minority-Class Recall

Training data volume, against each model’s complexity, explains why S and F recall differs so much. The training set holds only 2,037 S beats and 409 F beats, against 63,594 N and 5,025 V beats. It’s a severe imbalance.

Logistic Regression: Logistic Regression has only 72 coefficients across four classes. A model this simple needs little data to find a usable pattern, reaching an S-type recall of 0.9309 from only 2,037 examples.

Random Forest: Random Forest’s 300 trees add flexibility, but each tree’s prediction is averaged with 299 others. This helps common classes but dilutes a rare pattern like S rather than sharpening it. This is why Random Forest’s S-type recall, at 0.1198, is far lower than Logistic Regression’s.

CNN: The CNN has by far the most capacity, at 19,524 parameters. Relative to only 409 F and 2,037 S beats, this is far too many parameters for too little data. It has enough capacity to memorise the few examples it sees, without generalising to new ones. It is the direct case of overfitting on the minority classes specifically. Its recall on N and V, at 0.9967 and 0.8314, both the highest of the three models. It confirms that the network itself is not the problem where enough data exists.

The CNN’s failure on S and F is a consequence of too little training data, not a permanent limitation of the architecture. With substantially more S and F examples, the CNN would be the strongest candidate to also lead on minority-class recall.

3.6 Comparison of Model Characteristics

Table 11: Qualitative deployment comparison across the three models.

Criterion	Logistic Regression	Random Forest	CNN
Model size	Smallest	Largest	Intermediate
Interpretability	Higher	Moderate	Lower
Main strength	High S recall	Highest point-estimate macro F1	Highest accuracy and strong N/V
Main limitation	Many false abnormal alerts	Weak S/F recall	Zero S/F recall
Main clinical risk	False abnormal alerts	Missed S/F beats	Fails to detect all S/F cases

As Table 11 summarises, no single model wins on every front. Logistic Regression has the highest recall for abnormal beats. Random Forest has the highest F1. The CNN has the highest accuracy and the strongest recall on the classes it was given enough data to learn. If the priority is catching as many abnormal beats as possible, Logistic Regression performs strongest here. If

the priority shifts toward reducing false alarms once enough minority-class data exists, the CNN becomes the more promising long-term option. And if we can provide enough volume of data for S and F, CNN could be the unbeatable.

Task 4: Ethical Considerations and Discussion

Reflection on preprocessing and model selection decisions

1. **Patient-independent split:** The whole records were split, not individual beats. Beats from the same person are highly correlated. If it split by beats, it would let a model learn to recognise a patient, not a genuine arrhythmia pattern.
2. **Three separate model tracks:** Logistic Regression and Random Forest used the same 18 engineered features. The CNN used the raw waveform directly. This domain has decades of cardiology knowledge already encoded as features. This design tests whether a linear model suffices with good features, and whether a network can learn something better on its own.
3. **Per-class metric and prioritizing recall over accuracy:** Every result was broken down by class. A model's failure on one class, here F, is a patient-safety question. Again, being concern with patients' safety, recall is considered the primary metric rather than accuracy. Thus, accuracy and ROC-AUC is not focused in this work.
4. **Compliance:** This project complies with GDPR's anonymization principle. GDPR states that fully anonymized data, data that cannot be traced back to a real person, falls outside its scope entirely. This work only used the existing, already anonymized MIT-BIH dataset.

Model comparison and real-world applications

1. **Logistic Regression:** This model has the highest recall of the three, at 0.6115. Recall means how many real problems it actually catches. This matters most in a screening role, where missing a real abnormality is worse than a false alarm. This model also needs very little computing power. It works on simple, cheap devices, not just powerful computers. A real-world example is a basic wearable device, like a consumer smartwatch. Another example is a small rural clinic that cannot afford expensive medical equipment. Because recall is its biggest strength, it fits these low-resource, screening-first settings well.
2. **Random Forest:** This model has the highest F1 score of the three. F1 balances catching real problems against avoiding false alarms. This balance suits a setting where a human reviews the results afterward. A real-world example is a triage nurse reviewing daily ECG reports after the flag. This gives the reviewer a clear reason behind every decision.
3. **CNN:** This model has the highest recall for the two common heartbeat types, N and V. This makes it very reliable for spotting normal beats and one major abnormal type quickly. A real-world example is an emergency military recruitment where they try to avoid abnormal heart beats candidates. It could easily identify the major abnormal beat V over normal N ignoring the rarest S and F beats.

Ethical concerns

1. **Historical dataset:** Recorded between 1975 and 1979, at a single hospital.
2. **Small subject count:** Around 110,000 beats come from only 47 subjects where 1 subject is repeated once. Beat count is not subject count.
3. **Selection bias:** 23 recordings were random, 25 were deliberately enriched for rare arrhythmias. The enriched group supplies almost all S, V, and F evidence. Reported proportions reflect this design, not real-world prevalence. Mistaking these proportions for real prevalence would distort any risk assessment built on them.
4. **No demographic data available:** MIT-BIH records no age, sex, or ethnicity data. A demographic fairness analysis cannot be performed on this dataset, by any method. This is a hard limit of the data.

5. **Clinical consequences of specific errors:** Logistic Regression produced 4,322 false alarms and missed 127 abnormal beats. The CNN produced 44 false alarms and missed 664. A missed beat risks an untreated condition. Frequent false alarms risk alarm fatigue, where staff begin ignoring the system.

Solutions to mitigate bias and improve fairness

1. **Human oversight:** Model output should serve as a second opinion, never an automatic diagnosis, particularly for S and F.
2. **External validation:** The data is old, single-hospital and demographically unknown. Real deployment needs testing on newer, more diverse data first.
3. **Ongoing monitoring:** The training data is decades old. A deployed system needs monitoring for drift from this historical distribution, to catch silent performance loss early.
4. **Dataset enrichment:** Future versions of this dataset could collect more S and F examples specifically. Both classes suffer from too few examples to learn. More S and F data would let all three models be fairly retrained and retested on these classes. Also, recording demographic details such as sex could also help. It would add extra information about each existing beat.

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